

4(a)	any three from: in Venus fly traps 1 touch two (sensory hairs) ; 2 protons, leave hinge cells / enter hinge cell walls ; 3 Ca^{2+} ions move into (hinge) cells ; 4 water moves into (hinge) cells ; 5 (hinge) cells, become turgid / swell / increase in volume / expand ;	3
4(b)	any two from: 1 less / decreased, surface area (of leaf exposed) ; 2 less light absorbed by, chloroplasts / thylakoid membranes / grana / pigments / chlorophyll / light harvesting complexes / photosystems ; 3 fewer stomata, exposed / available (to air) ; 4 less carbon dioxide, enters / absorbed ;	2
4(c)	any six from: <i>similarities</i> 1 photoactivation of chlorophyll ; 2 (energetic) electrons move, along / down, ETC ; 3 chemiosmosis ; 4 ATP produced ; <i>differences</i> 5 PSI and PSII in non-cyclic and only PSI in cyclic ; 6 photolysis / oxygen produced, in non-cyclic only ; ORA 7 reduced NADP in non-cyclic only ; ORA 8 electrons return to, same photosystem / PSI, in cyclic ;	6